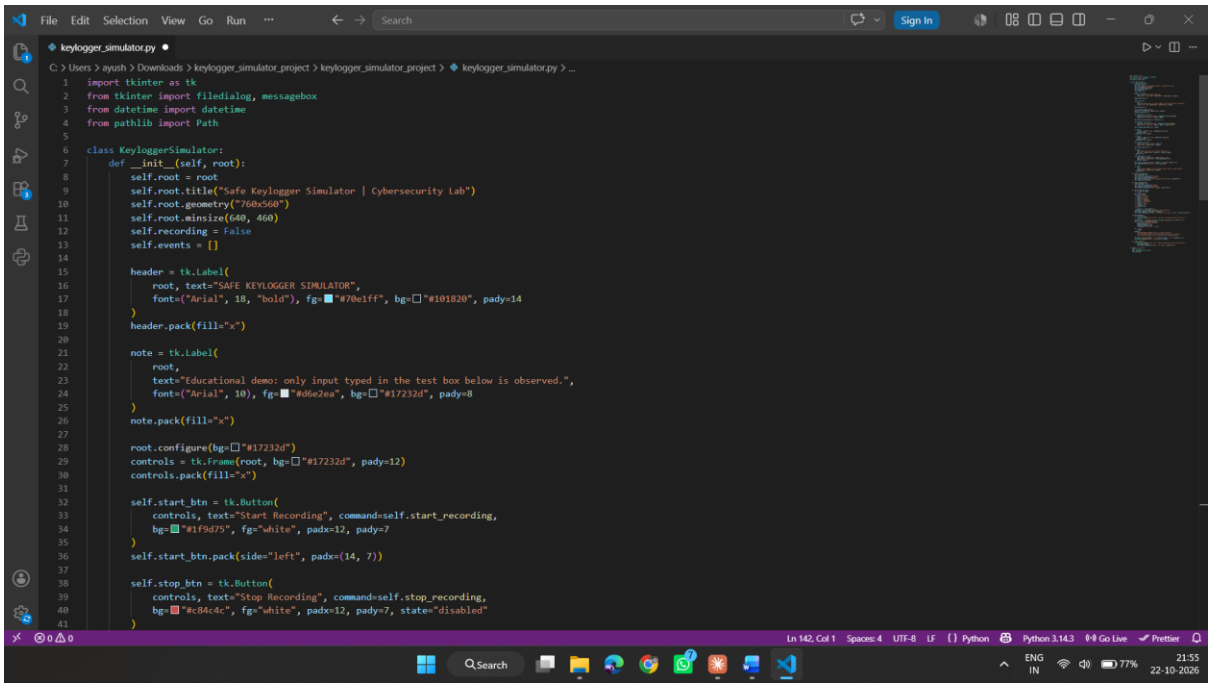
Topic: Keylogger Simulator

Step-1: I entered the code, it described that I makes a window with Start, Stop, Save, and Clear buttons plus a text box. When you hit Start, it records every key you type in that box, each with a timestamp, into a list. Save Log writes those keystrokes to a .txt file, and Clear wipes them.



```python
import tkinter as tk
from tkinter import filedialog, messagebox
from datetime import datetime
from pathlib import Path

class KeyloggerSimulator:
    def __init__(self, root):
        self.root = root
        self.root.title("Safe Keylogger Simulator | Cybersecurity Lab")
        self.root.geometry("760x560")
        self.root.minsize(640, 460)
        self.recording = False
        self.events = []

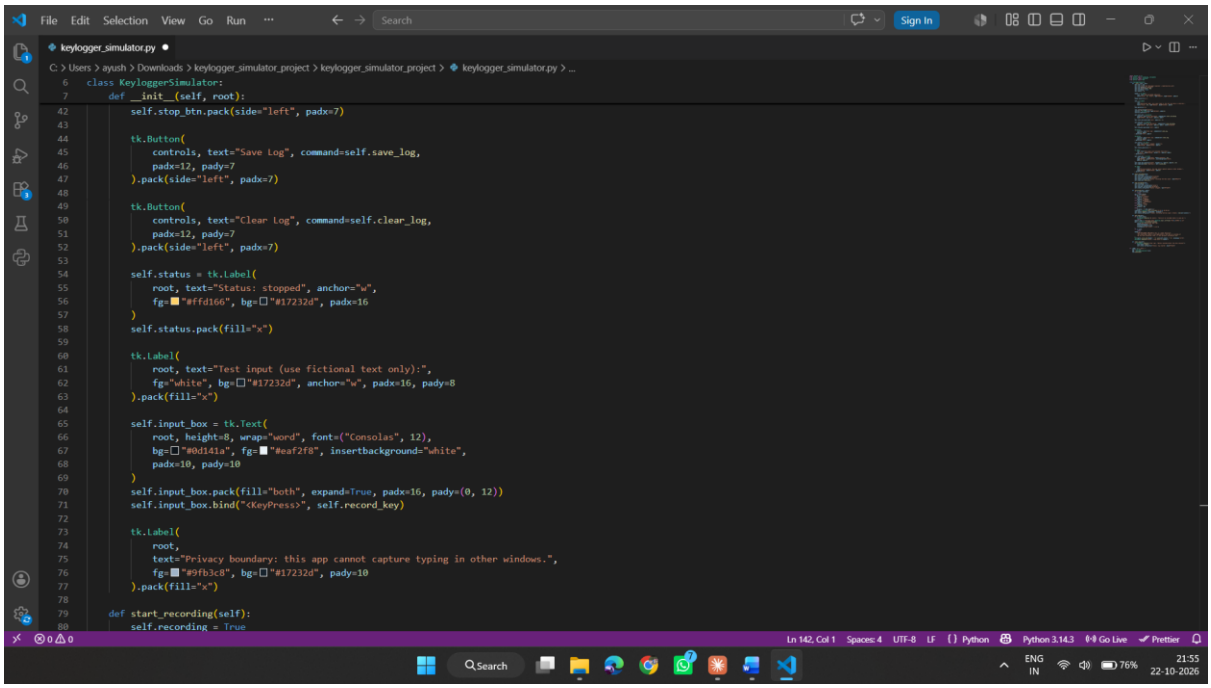
        header = tk.Label(
            root, text="SAFE KEYLOGGER SIMULATOR",
            font=("Arial", 18, "bold"), fg="#70e1ff", bg="#101820", pady=14
        )
        header.pack(fill="x")

        note = tk.Label(
            root,
            text="Educational demo: only input typed in the test box below is observed.",
            font=("Arial", 10), fg="#d6e2ea", bg="#17232d", pady=8
        )
        note.pack(fill="x")

        root.configure(bg="#17232d")
        controls = tk.Frame(root, bg="#17232d", pady=12)
        controls.pack(fill="x")

        self.start_btn = tk.Button(
            controls, text="Start Recording", command=self.start_recording,
            bg="#1f9d75", fg="white", padx=12, pady=7
        )
        self.start_btn.pack(side="left", padx=(14, 7))

        self.stop_btn = tk.Button(
            controls, text="Stop Recording", command=self.stop_recording,
            bg="#c84c4c", fg="white", padx=12, pady=7, state="disabled"
        )
```



```python
class KeyloggerSimulator:
    def __init__(self, root):
        self.stop_btn.pack(side="left", padx=7)

        tk.Button(
            controls, text="Save Log", command=self.save_log,
            padx=12, pady=7
        ).pack(side="left", padx=7)

        tk.Button(
            controls, text="Clear Log", command=self.clear_log,
            padx=12, pady=7
        ).pack(side="left", padx=7)

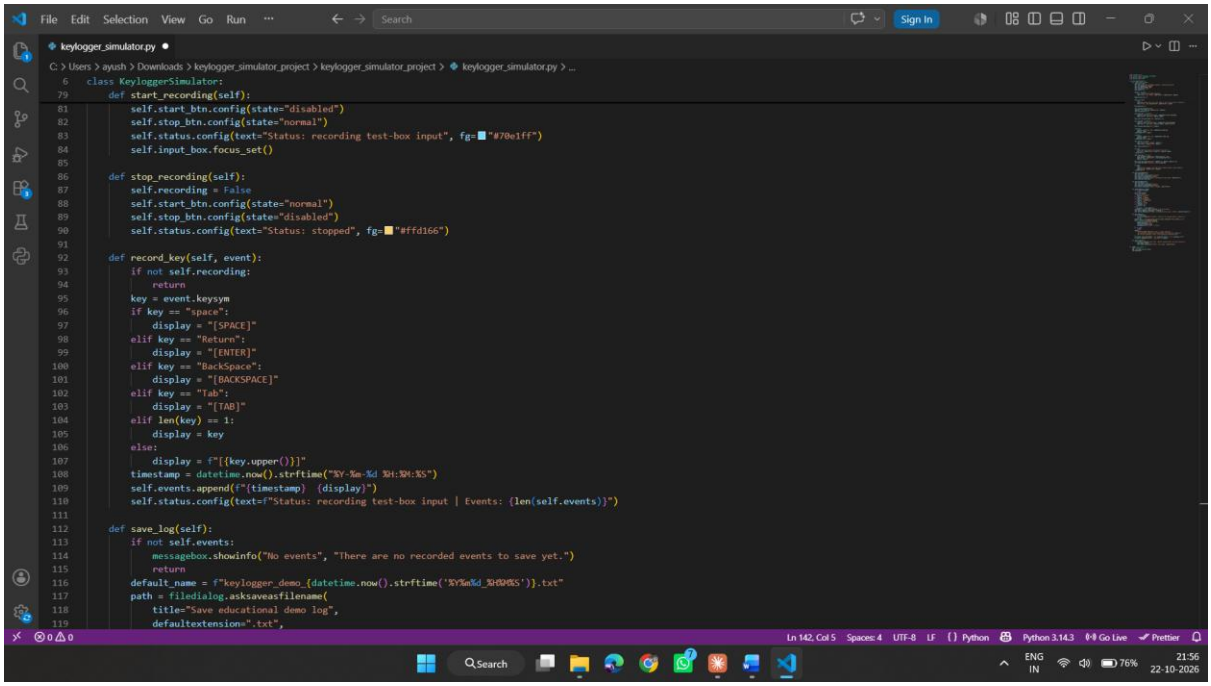
        self.status = tk.Label(
            root, text="Status: stopped", anchor="w",
            fg="#ffd166", bg="#17232d", padx=16
        )
        self.status.pack(fill="x")

        tk.Label(
            root, text="Test input (use fictional text only):",
            fg="white", bg="#17232d", anchor="w", padx=16, pady=8
        ).pack(fill="x")

        self.input_box = tk.Text(
            root, height=8, wrap="word", font=("Consolas", 12),
            bg="#0d141a", fg="#eaf2f8", insertbackground="white",
            padx=10, pady=10
        )
        self.input_box.pack(fill="both", expand=True, padx=16, pady=(0, 12))
        self.input_box.bind("<KeyPress>", self.record_key)

        tk.Label(
            root,
            text="Privacy boundary: this app cannot capture typing in other windows.",
            fg="#9fb3c8", bg="#17232d", pady=10
        ).pack(fill="x")

    def start_recording(self):
        self.recording = True
```

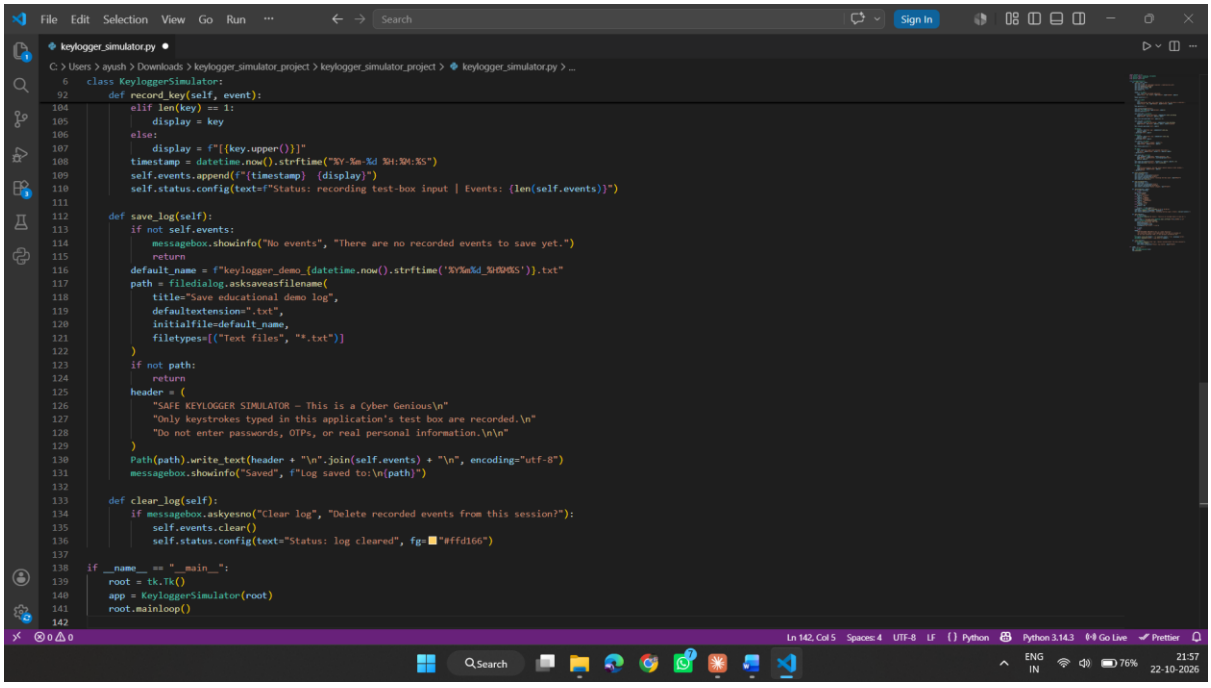


```
class KeyloggerSimulator:
    def start_recording(self):
        self.start_btn.config(state="disabled")
        self.stop_btn.config(state="normal")
        self.status.config(text="Status: recording test-box input", fg="#70e1ff")
        self.input_box.focus_set()

    def stop_recording(self):
        self.recording = False
        self.start_btn.config(state="normal")
        self.stop_btn.config(state="disabled")
        self.status.config(text="Status: stopped", fg="#ffd166")

    def record_key(self, event):
        if not self.recording:
            return
        key = event.keysym
        if key == "space":
            display = "[SPACE]"
        elif key == "Return":
            display = "[ENTER]"
        elif key == "BackSpace":
            display = "[BACKSPACE]"
        elif key == "Tab":
            display = "[TAB]"
        elif len(key) == 1:
            display = key
        else:
            display = f"[{key.upper()}]"
        timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        self.events.append(f"{timestamp}  {display}")
        self.status.config(text=f"Status: recording test-box input | Events: {len(self.events)}")

    def save_log(self):
        if not self.events:
            messagebox.showinfo("No events", "There are no recorded events to save yet.")
            return
        default_name = f"keylogger_demo_{datetime.now().strftime('%Y%m%d_%H%M%S')}.txt"
        path = filedialog.asksaveasfilename(
            title="Save educational demo log",
            defaultextension=".txt",
```



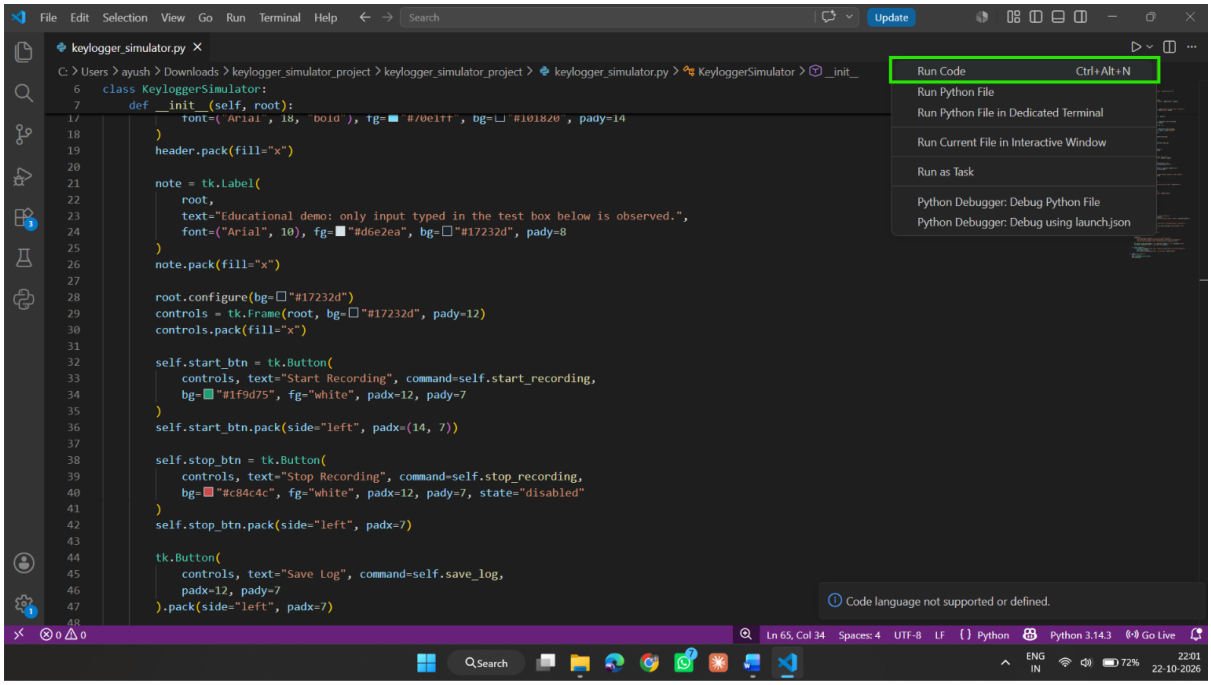
```
class KeyloggerSimulator:
    def record_key(self, event):
        elif len(key) == 1:
            display = key
        else:
            display = f"[{key.upper()}]"
        timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        self.events.append(f"{timestamp}  {display}")
        self.status.config(text=f"Status: recording test-box input | Events: {len(self.events)}")

    def save_log(self):
        if not self.events:
            messagebox.showinfo("No events", "There are no recorded events to save yet.")
            return
        default_name = f"keylogger_demo_{datetime.now().strftime('%Y%m%d_%H%M%S')}.txt"
        path = filedialog.asksaveasfilename(
            title="Save educational demo log",
            defaultextension=".txt",
            initialfile=default_name,
            filetypes=[("Text files", "*.txt")]
        )
        if not path:
            return
        header = (
            "SAFE KEYLOGGER SIMULATOR — This is a Cyber Genious\n"
            "Only keystrokes typed in this application's test box are recorded.\n"
            "Do not enter passwords, OTPs, or real personal information.\n\n"
        )
        Path(path).write_text(header + "\n".join(self.events) + "\n", encoding="utf-8")
        messagebox.showinfo("Saved", f"Log saved to:\n{path}")

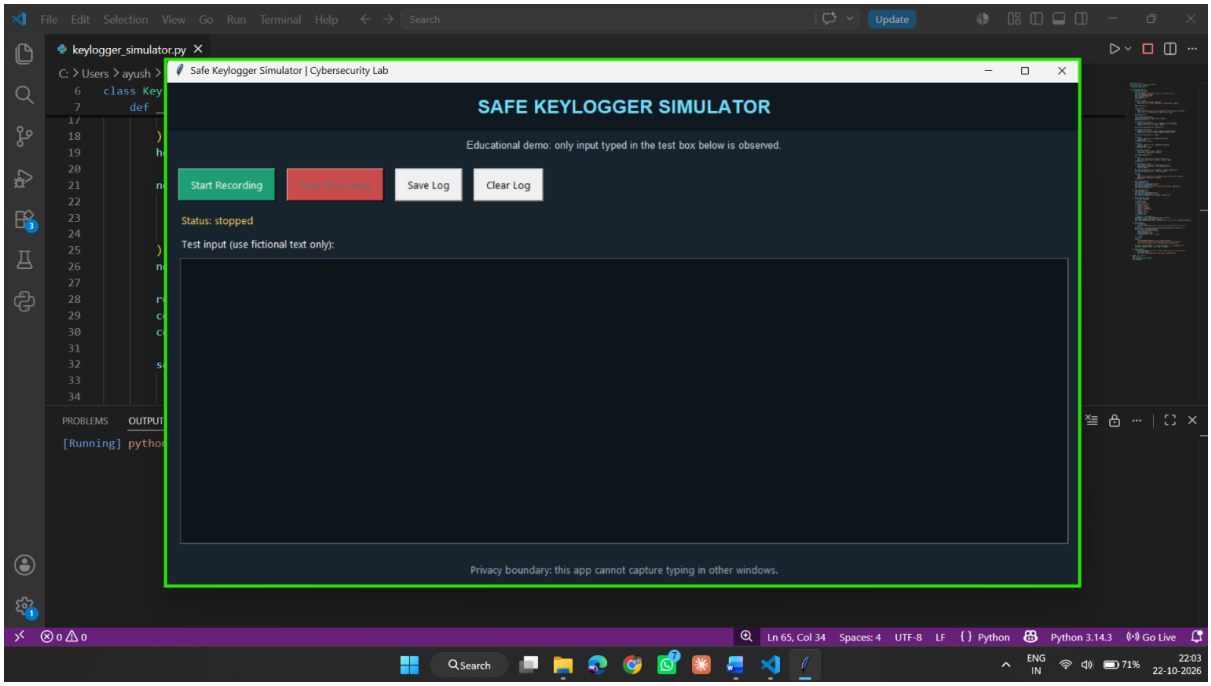
    def clear_log(self):
        if messagebox.askyesno("Clear log", "Delete recorded events from this session?"):
            self.events.clear()
            self.status.config(text="Status: log cleared", fg="#ffd166")

if __name__ == "__main__":
    root = tk.Tk()
    app = KeyloggerSimulator(root)
    root.mainloop()
```

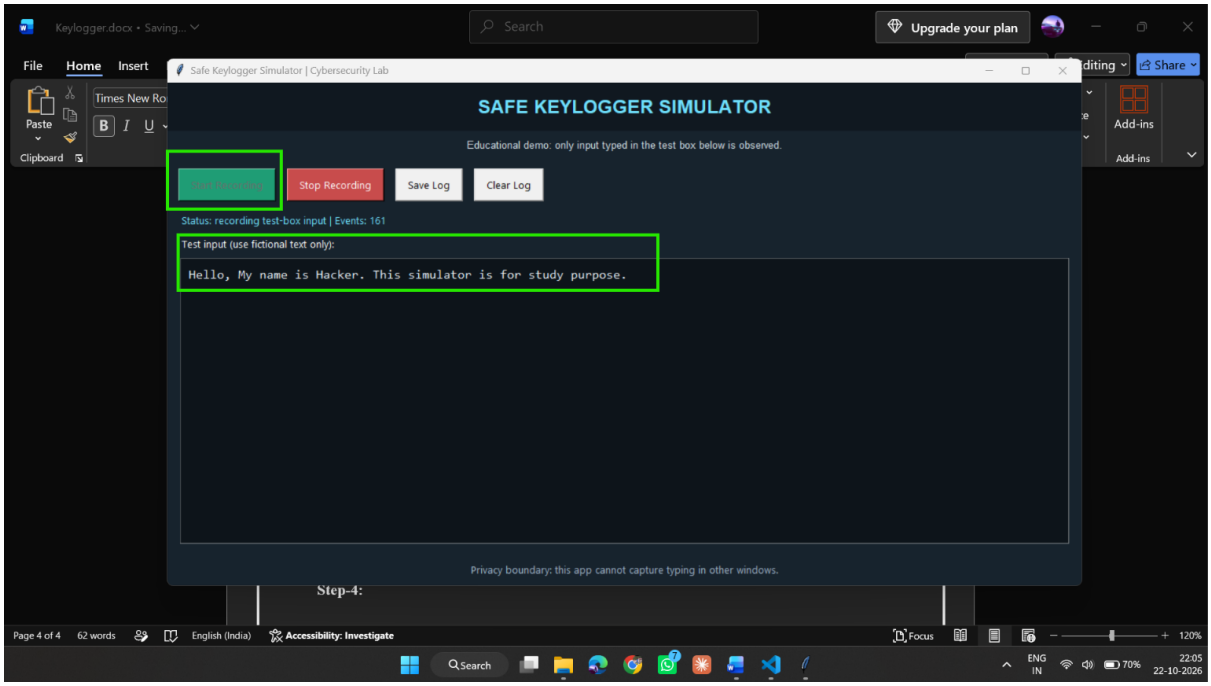
Step-2: I clicked on run code to execute the keylogger.



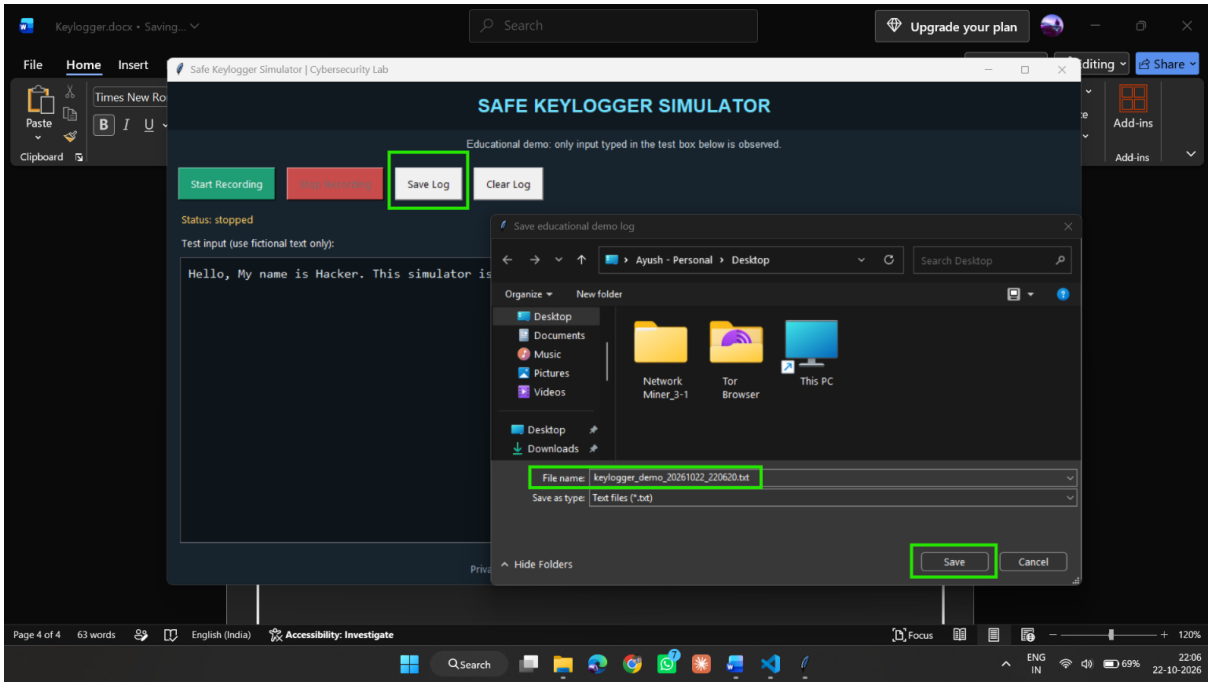
Step-3: This is the home page of the Keylogger.



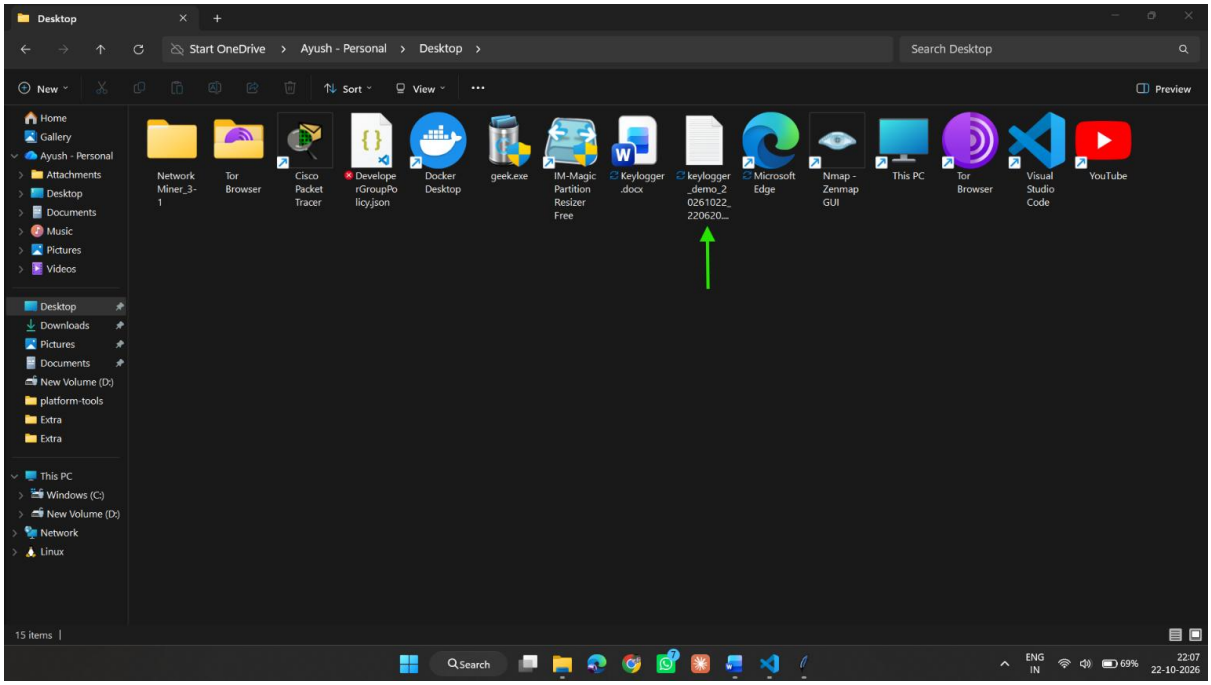
Step-4: Now I click on the Start recording and I enter the text inside the field.



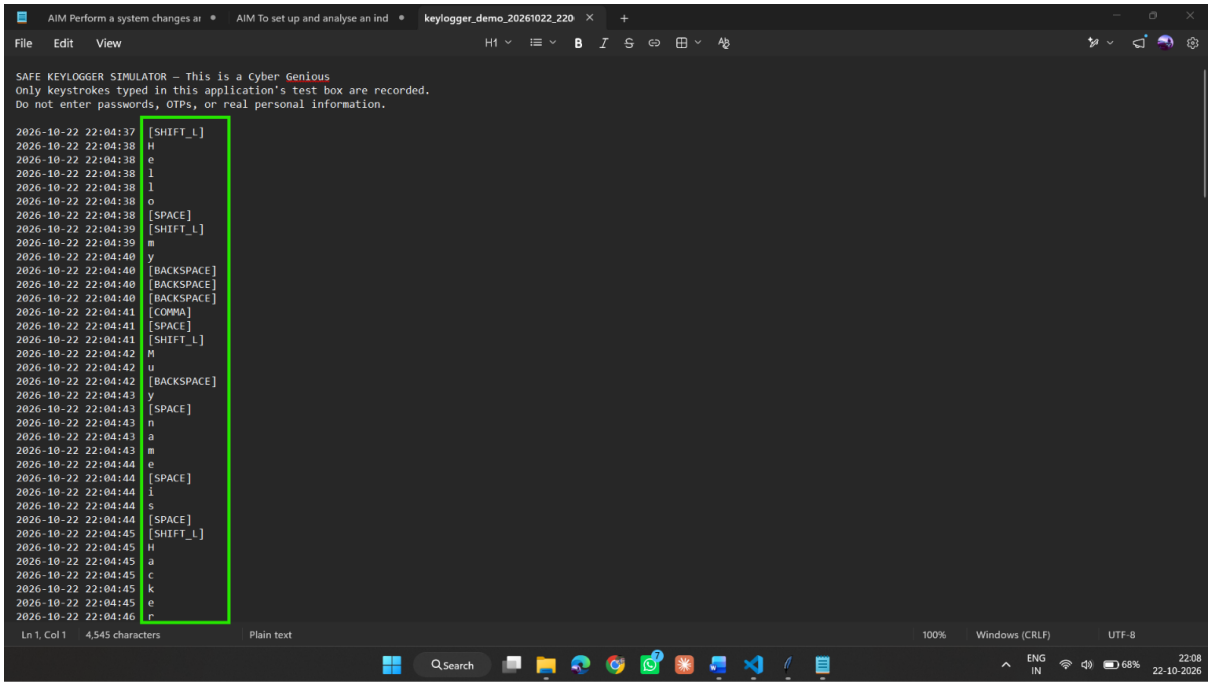
Step-5: Now I click on the Save log and then saved it on desktop.

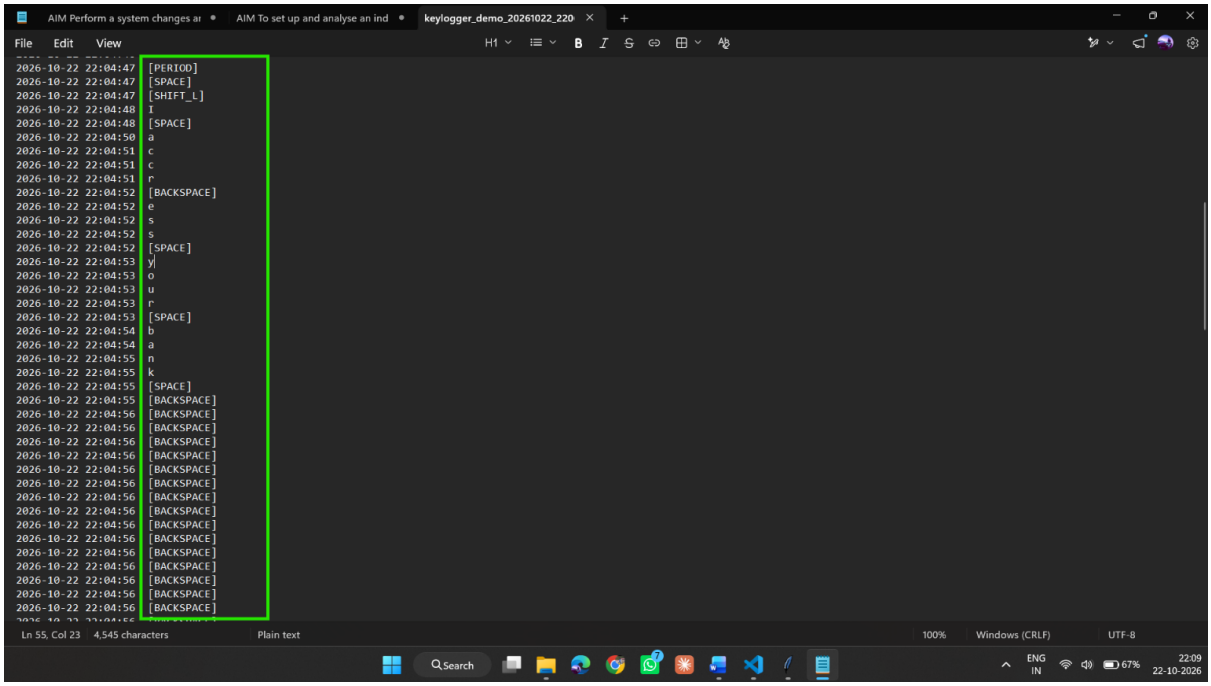


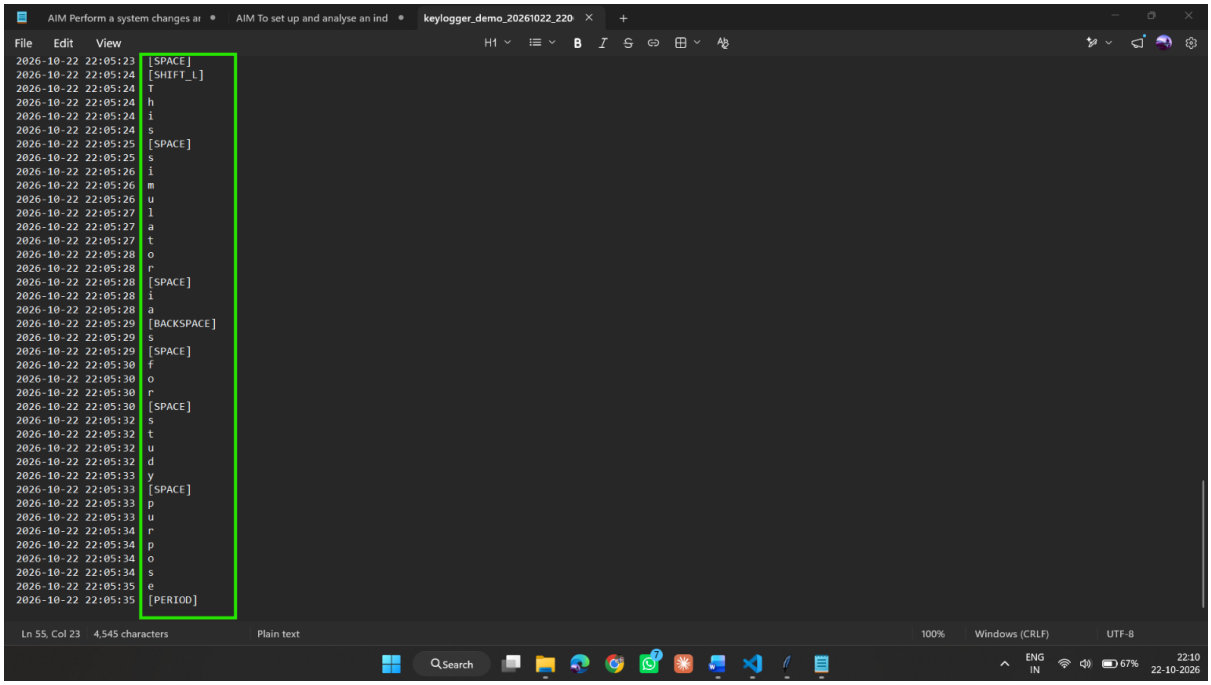
Step-6: File that can be saved on desktop.



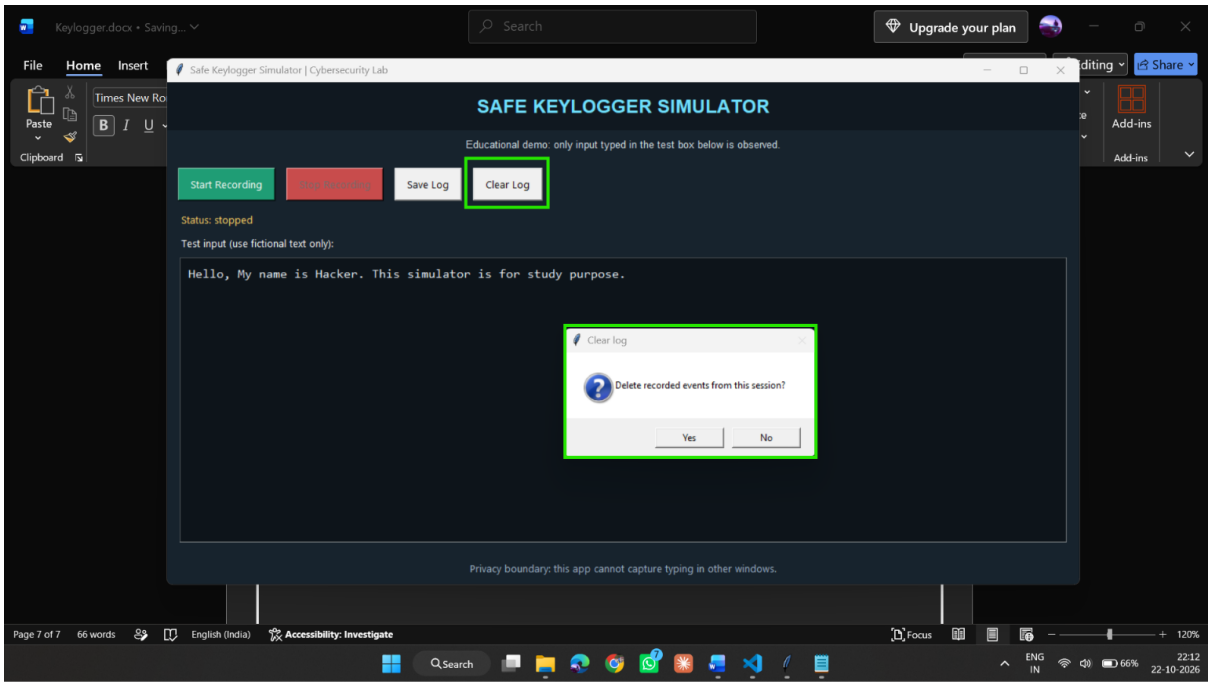
Step-7: When I clicked on that file all the text which I entered that can be saved inside that text file. How many time I press the backspace, how many time I press the spacebar all the things can be recorded.




AIM Perform a system changes ar
AIM To set up and analyse an ind
keylogger_demo_20261022_220
File Edit View
2026-10-22 22:04:47 [PERIOD]
2026-10-22 22:04:47 [SPACE]
2026-10-22 22:04:47 [SHIFT_L]
2026-10-22 22:04:48 I
2026-10-22 22:04:48 [SPACE]
2026-10-22 22:04:50 a
2026-10-22 22:04:51 c
2026-10-22 22:04:51 c
2026-10-22 22:04:51 r
2026-10-22 22:04:52 [BACKSPACE]
2026-10-22 22:04:52 e
2026-10-22 22:04:52 s
2026-10-22 22:04:52 s
2026-10-22 22:04:52 [SPACE]
2026-10-22 22:04:53 y
2026-10-22 22:04:53 o
2026-10-22 22:04:53 u
2026-10-22 22:04:53 r
2026-10-22 22:04:53 [SPACE]
2026-10-22 22:04:54 b
2026-10-22 22:04:54 a
2026-10-22 22:04:55 n
2026-10-22 22:04:55 k
2026-10-22 22:04:55 [SPACE]
2026-10-22 22:04:55 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
2026-10-22 22:04:56 [BACKSPACE]
Ln 55, Col 23 4,545 characters
Plain text
100%
Windows (CRLF)
UTF-8
Search
ENG IN
67%
22:09
22-10-2026


AIM Perform a system changes ar
AIM To set up and analyse an ind
keylogger_demo_20261022_220
File Edit View
2026-10-22 22:05:23 [SPACE]
2026-10-22 22:05:24 [SHIFT_L]
2026-10-22 22:05:24 T
2026-10-22 22:05:24 h
2026-10-22 22:05:24 i
2026-10-22 22:05:24 s
2026-10-22 22:05:25 [SPACE]
2026-10-22 22:05:25 s
2026-10-22 22:05:26 i
2026-10-22 22:05:26 m
2026-10-22 22:05:26 u
2026-10-22 22:05:27 l
2026-10-22 22:05:27 a
2026-10-22 22:05:27 t
2026-10-22 22:05:28 o
2026-10-22 22:05:28 r
2026-10-22 22:05:28 [SPACE]
2026-10-22 22:05:28 i
2026-10-22 22:05:28 a
2026-10-22 22:05:29 [BACKSPACE]
2026-10-22 22:05:29 s
2026-10-22 22:05:29 [SPACE]
2026-10-22 22:05:30 f
2026-10-22 22:05:30 o
2026-10-22 22:05:30 r
2026-10-22 22:05:30 [SPACE]
2026-10-22 22:05:32 s
2026-10-22 22:05:32 t
2026-10-22 22:05:32 u
2026-10-22 22:05:32 d
2026-10-22 22:05:33 y
2026-10-22 22:05:33 [SPACE]
2026-10-22 22:05:33 p
2026-10-22 22:05:33 u
2026-10-22 22:05:34 r
2026-10-22 22:05:34 p
2026-10-22 22:05:34 o
2026-10-22 22:05:34 s
2026-10-22 22:05:35 e
2026-10-22 22:05:35 [PERIOD]
Ln 55, Col 23 4,545 characters
Plain text
100%
Windows (CRLF)
UTF-8
Search
ENG IN
67%
22:10
22-10-2026

Step-8: I can also clear the logs of the file which I created. When I click on the clear logs and press yes it will clear all the logs.



Conclusion: I understand how the keylogger work. How keylogger can save the data inside the text file. If an unauthorized person can enter into your system he might be access your sensitive credentials of the system.